Satwave Arrays

Satwave Flat Panel AESA

Optimized Design, Maximized Performance

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THE PROBLEM

The market need for flat panel antennas has exploded with the deployment of thousands of Low-Earth Orbit (LEO) satellites to support the growing global demand for data for applications such as Land Mobility, Aviation, Maritime and particularly, Defense and Government.

As a result, the development of flat-panel Active Electronically Scanned Array (AESA) antennas has shifted from being "niche technology" to becoming "critical infrastructure" necessary to communicate most effectively with LEO satellite constellations and provide seamless mobility.

SATWAVE ARRAYS SOLUTION

Satwave Arrays is bringing to market two **Beam-Forming Integrated Circuit (BFIC) AESAs** to address that unmet need for high-performance, efficient and viable AESAs. Satwave has built reliable, robust, and high-quality Ku-Band and Ka-Band systems that support mobility and fixed connectivity across LEO, MEO, and GEO.

We have validated the arrays through extensive measurements and tests in a Planar Near Field and Compact Antenna Test ranges yielding results that demonstrate strong performance.

Ku-Band AESA

Ka-Band AESA

ABOUT SATWAVE ARRAYS, INC.

Satwave Arrays is a satellite antenna company based in the city of Atlanta, USA. We design, develop and deliver Active Electronically Scanned Array antennas for satellite communications. We are building reliable, robust and high quality phased-array systems with the ability to support mobility and fixed connectivity with satellites on any orbit (LEO, MEO and GEO) and in any constellation.

Satwave’s focus is on military and government applications as well as commercial markets. We started operations in mid-2023 and have a talented team of engineers and experienced industry veterans.

WHY SATWAVE AESAs?

- **Electronic Beam Steering**
  Unlike traditional parabolic dishes, phased arrays have no moving parts.
- **Reliability**
  No motors or gears means significantly lower maintenance costs and higher "uptime" in harsh environments.
- **Portability**
  Flat panel antennas are easier to transport (even man-packable).
- **Upgradability**
  Satwave AESAs are designed to be upgraded to support higher security levels and other features based on customer requirements.
- **Agility**
  Ability to switch "locks" between LEO satellites in microseconds as they zip across the sky in minutes.
- **Low Profile**
  Bulky dishes are prone to wind damage and vibration. Satwave AESAs are "low-impact" and can be integrated directly into the roof or mast.
- **Resilience**
  Being phased array antennas, Satwave AESAs would be more resistant to jamming than traditional antennas.

Benefits of Satwave’s Approach

- AESA design scalable to custom sizes and multi-panel antennas.
- Proprietary, ML optimized, inner layers for wideband performance, enhanced cross-polarization, and improved efficiency at large scan angles.
- PCB-based design, manufacturable at lower cost at scale.
- Built on commercially available analog BFICs.
- Targeted to be IP67 compliant and to operate in MIL-STD-810H environments.
- Full system - Antenna Control Unit and Auto tracking algorithm to integrate with external modem available in the market.
- Driven by FreeRTOS software (supported by Amazon and used by millions of devices).

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Enabling Mobility

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